

LINUX COMMANDS EXPLANATION

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This document explains the Linux commands used in the Linux Admin Dashboard project and their purpose.

1. clear

Purpose:

Clears the terminal screen.

Usage in Project:

Used before displaying menus and submenus to keep the interface clean and easy to read.

2. hostname

Purpose:

Displays the hostname (computer name) of the system.

Usage in Project:

Used in the System Information module and Report Generation module.

3. whoami

Purpose:

Displays the username of the currently logged-in user.

Usage in Project:

Used in the System Information module and Report Generation module.

4. uptime -p

Purpose:

Shows how long the system has been running.

Usage in Project:

Used in the System Information module and Report Generation module.

5. free -h

Purpose:

Displays memory (RAM) usage in a human-readable format.

Usage in Project:

Used in the System Information module and Report Generation module.

6. df -h

Purpose:

Displays disk usage information in a human-readable format.

Usage in Project:

Used in the System Information module and Report Generation module.

7. cut

Purpose:

Extracts specific fields from text files.

Usage in Project:

Used to extract usernames from the /etc/passwd file.

Example:

```
cut -d: -f1 /etc/passwd
```

8. groups

Purpose:

Displays the groups that the current user belongs to.

Usage in Project:

Used in the Users and Groups module.

9. who

Purpose:

Displays users currently logged into the system.

Usage in Project:
Used in the Users and Groups module.

10. find

Purpose:
Searches for files and directories.

Usage in Project:
Used to search for files and to find files with 777 permissions.

Examples:
`find ~ -name "filename"`
`find . -type f -perm 777`

11. wc -l

Purpose:
Counts the number of lines in the output.

Usage in Project:
Used to count users and files.

Example:
`cut -d: -f1 /etc/passwd | wc -l`

12. du -sh

Purpose:
Displays the total size of a directory.

Usage in Project:
Used in the File Management module.

13. touch

Purpose:
Creates a new empty file.

Usage in Project:
Used in the File Management module to create files.

14. rm -i

Purpose:
Deletes files after asking for confirmation.

Usage in Project:
Used in the File Management module to remove files safely.

15. ls -l

Purpose:
Displays detailed file information including permissions, owner, size, and date.

Usage in Project:
Used in the Permission Audit module.

16. tail

Purpose:
Displays the last lines of a file.

Usage in Project:
Used to display the latest entries from the system log file.

Example:
`tail -20 /var/log/syslog`

17. grep

Purpose:
Searches for specific text inside files.

Usage in Project:
Used to search for keywords in the system log file.

Example:

```
grep -i "error" /var/log/syslog
```

18. sudo

Purpose:

Runs commands with administrator privileges.

Usage in Project:

Used to access protected system log files.

19. tar -czf

Purpose:

Creates compressed archive files.

Usage in Project:

Used in the Backup System module to create backups.

Example:

```
tar -czf backup.tar.gz directory_name
```

20. date

Purpose:

Displays the current date and time.

Usage in Project:

Used for report generation and backup file naming.

Example:

```
date
```

```
date +%Y%m%d_%H%M%S
```

21. chmod

Purpose:

Changes file permissions.

Usage in Project:

Used during project setup to make the script executable.

Example:

```
chmod +x linux_dashboard.sh
```

DEVELOPMENT COMMANDS

The following commands were used during development and testing of the project:

1. nano
 - Used to create and edit files.
2. mkdir
 - Used to create project folders.
3. mv
 - Used to move files into the project directory.
4. ls
 - Used to list files and folders.
5. cd
 - Used to navigate between directories.
6. pwd
 - Used to display the current directory.
7. bash -n
 - Used to check the script for syntax errors.